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Original Research Article

In-vitro susceptibility of cefiderocol and other comparators in carbapenem-resistant Gram-negative bacilli: A study from India

Akansha Didwania [a], Sarita Mohapatra [b,*], Deepak Kocher [b], Hitender Gautam [b], Priyanka Kumari [b], Arvind Kumar [c], Manish Soneja [c], Naval K. Vikram [c], Seema Sood [b], Benu Dhawan [b], Bimal Kumar Das [b], Rama Chaudhry [b], Manoranjan Mahapatra [d], Naveet Wig [c], Arti Kapil [b]

[a] *Infectious Diseases, Department of Medicine and Microbiology, All India Institute of Medical Sciences, New Delhi, India*
[b] *Department of Microbiology, All India Institute of Medical Sciences, New Delhi, India*
[c] *Department of Medicine, All India Institute of Medical Sciences, New Delhi, India*
[d] *Department of Haematology, All India Institute of Medical Sciences, New Delhi, India*

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ABSTRACT

Introduction: Cefiderocol is a parenteral catechol-type siderophore cephalosporin, which has been approved for the treatment of Gram-negative bacterial infections. Its activity among the carbapenem-resistant Gram-negative bacilli (CR-GNBs) in India is largely unknown.
Methodology: We tested *in-vitro* susceptibility of cefiderocol in 84 CR-GNB [carbapenem-resistant *Acinetobacter baumannii* (CRAB), carbapenem-resistant *Pseudomonas aeruginosa* (CRPA), carbapenem-resistant *Escherichia coli* (CREC) and carbapenem-resistant *Klebsiella pneumoniae* (CRKP)] by broth microdilution (BMD) and disc diffusion (DD) using Clinical and Laboratory Standards Institute (CLSI) and European Committee on Antimicrobial Susceptibility Testing (EUCAST) breakpoints and concordance of DD was compared with BMD. Cefiderocol susceptibility was evaluated based on the presence of bla_{NDM} and $bla_{\mathrm{OXA\text{-}48}}$ genes.
Results: Following CLSI and EUCAST breakpoints, 76.2% [CRKP: 95.65% (22/23); CRPA: 80% (16/20); CREC: 65% (13/20); CRAB: 61.9% (13/21)] and 52.3% [CRKP: 82.6% (19/23); CRPA: 50% (10/20); CREC: 20% (4/20)] of CR-GNBs were observed susceptible to cefiderocol, respectively. The susceptibility to cefiderocol was seen higher with lower MIC_{50} and MIC_{90} values for CRKP [1 μg/ml; 4 μg/ml] and CRPA [2 μg/ml; 8 μg/ml] in comparison to the other two groups of organisms. The categorical agreement between BMD and DD was found between 90% and 100% for all organisms. Cefiderocol susceptibility among the isolates harbouring bla_{NDM}, $bla_{\mathrm{OXA\text{-}48}}$, and both bla_{NDM} and $bla_{\mathrm{OXA\text{-}48}}$ genes was observed 75% (24/32); 100% (3/3); and 66.6% (4/6), respectively.
Conclusion: Cefiderocol demonstrated high *in-vitro* activity against CRKP and CRPA isolates from India with lower MIC_{50} and MIC_{90} irrespective of the presence of bla_{NDM} and $bla_{\mathrm{OXA\text{-}48}}$ genes.

1. Introduction

Gram-negative bacilli (GNB) are the most common pathogens causing infections in hospital and community settings, worldwide. There is an urgent need for newer antibiotics due to the increasing prevalence of carbapenem-resistant GNBs (CR-GNBs) i.e. carbapenem-resistant *Acinetobacter baumannii* (CRAB), carbapenem-resistant *Pseudomonas aeruginosa* (CRPA), and carbapenem-resistant Enterobacterales (CRE) (includes carbapenem-resistant *Escherichia coli* (CREC) and carbapenem-resistant *Klebsiella pneumoniae* (CRKP) [1]. The annual antimicrobial

* Corresponding author. Department of Microbiology, All India Institute of Medical Sciences, Room: 2058A, Second Floor Teaching Block, Ansari Nagar West, New Delhi, 110049, India.
E-mail addresses: akanshadidwania31@gmail.com (A. Didwania), drsarita2005@gmail.com (S. Mohapatra), deepakkocher.96@gmail.com (D. Kocher), drhitender@gmail.com (H. Gautam), aiims.priya21@gmail.com (P. Kumari), linktoarvind@gmail.com (A. Kumar), manishsoneja@gmail.com (M. Soneja), navalvikram@gmail.com (N.K. Vikram), seemalsood@gmail.com (S. Sood), dhawanb@gmail.com (B. Dhawan), tezpur.bimal@gmail.com (B.K. Das), drramach@gmail.com (R. Chaudhry), mrmahapatra@hotmail.com (M. Mahapatra), naveetwig@gmail.com (N. Wig).

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resistance research and surveillance network report from the year 2016 – 2021 by Indian Council of Medical Research (ICMR) showed a 15% – 20% decreased susceptibility to carbapenems in *E. coli*, *K. pneumoniae*, *A. baumannii*, and *P. aeruginosa* isolates [2].

The newer β-lactam/β-lactamase inhibitors (BL/BLI) have activity against extended-spectrum beta-lactamases (ESBLs), and carbapenemases without significant action against metallo-β-lactamase. (with variable activity against class A and class D carbapenemases). Cefiderocol is a novel catechol-type siderophore cephalosporin that forms a chelated complex with extracellular ferric ions. It enters the periplasmic space using bacterial iron transporters and inhibits cell wall synthesis by binding to penicillin-binding protein 3 (PBP3). Its unique "Trojan Horse" entry method and structure provide stability against beta-lactamases and carbapenemases, including metallo-β-lactamase. Cefiderocol also overcomes resistance from porin channel mutations and efflux pump overexpression [3]. Accurate testing conditions are required for the correct interpretation of its *in-vitro* susceptibility test results. Currently, the Clinical and Laboratory Standards Institute (CLSI) and the European Committee on Antimicrobial Susceptibility Testing (EUCAST) have approved broth microdilution (BMD) and disc diffusion for *in-vitro* testing of cefiderocol [4,5]. The current study aimed to compare the reproducibility of cefiderocol susceptibility results by disc diffusion test with that of BMD following CLSI and EUCAST guidelines among the CR-GNBs of Indian origin with various resistance profiles.

2. Materials and methods

A prospective observational study was conducted from 2021 to 2022. Ethical approval for study was obtained from the institute ethics committee [All India Institute of Medical Sciences, New Delhi (reference number: IEC-760/November 12, 2021,RP-31/2021)]. We included eighty-four non-duplicate clinically significant CR-GNB isolates from patients with clinical diagnosis of bloodstream infection (BSI), pneumonia and urinary tract infection (UTI) from a tertiary care institute, which caters to the northern part of India. The pathogens of interest were CRE (CREC, CRKP), CRPA, and CRAB. We tried to collect at least 20 isolates from each organism type. The organisms were identified using Matrix Assisted Laser Desorption Ionization-Time of Flight (MALDI-TOF) (Biomérieux, France) and antimicrobial susceptibility testing (AST) was performed by VITEK-2 (Biomérieux, France). Organisms found resistant to either imipenem or meropenem were considered carbapenem-resistant (as defined by CLSI) and further tested for cefiderocol susceptibility.

Cefiderocol AST was performed by broth microdilution using iron-depleted cation-adjusted Mueller Hinton broth (CA-MHB) following CLSI 2022 guidelines. Cefiderocol was tested in doubling dilutions ranging from 0.03 to 32 μg/ml. Each run included quality, drug, and sterility controls. For quality control, *Escherichia coli* ATCC 25922 and *Pseudomonas aeruginosa* ATCC 27853 strains were used. Each isolate was tested in duplicate on a single microtiter plate, and experiments were performed three times to ensure reproducibility. Results were interpreted by three independent observers to avoid bias, following both CLSI 2022 and EUCAST 2022 breakpoints. Enterobacterales and *P. aeruginosa* have defined breakpoints in both CLSI and EUCAST, while *A. baumannii* has breakpoints only in CLSI. MIC_{50} (Minimum inhibitory concentration of 50 percentile) and MIC_{90} (Minimum inhibitory concentration of 90 percentile) values were calculated for all organisms.

Disc diffusion was performed using 30μg cefiderocol discs (Liofilchem, Italy) on cation-adjusted Mueller Hinton agar (CA-MHA) plates. Zones of inhibition were read by three observers according to CLSI and EUCAST breakpoints. Both Enterobacterales and *P. aeruginosa* have defined breakpoints in CLSI and EUCAST. For *A. baumannii, f*ollowing CLSI guideline, a zone of inhibition more than 15 mm is defined as susceptible and any zone less than equal to 14 mm should be corroborated with the BMD test result, while EUCASTconsiders a zone over 17 mm as susceptible. EUCAST has defined Area of Technical Uncertainty (ATU) for Enterobacterales and *P. aeruginosa* in addition to susceptible and resistant category, which requires careful interpretation.

We tested *in-vitro* susceptibility of CR-GNB isolates to meropenem, colistin, and BL/BLIs (piperacillin-tazobactam, ceftolozane-tazobactam, ceftazidime-avibactam) using Sensititre AST plates (Thermo Scientific, USA). These 96-well microtiter plates are pre-loaded with two-fold serial dilutions of the drugs. A 0.5 McFarland suspension of isolated colonies was prepared in sterile water, then 10 μL was added to 11 mL of CAMHB from the kit. Each well received 50 μL of this suspension. Results were read after 16–20 h of incubation, and MICs were interpreted as per CLSI 2022 guidelines. All CR-GNBs were tested for carbapenemase genes (*bla_{NDM}* and *$bla_{\mathrm{OXA\text{-}48}}$*) using PCR with standard primers [6,7] We processed the samples for *bla_{NDM}* and *$bla_{\mathrm{OXA\text{-}48}}$* genes as these are the predominant carbapenem-resistant genes observed among the GNBs in India. For agreement analysis, disc diffusion was compared with the reference method (BMD). Categorical agreement (CA) was defined as number of isolates where BMD and disc diffusion results are in complete agreement. Minor error (mE) occurred when BMD results were susceptible/resistant and disc diffusion was intermediate or vice versa. Major error (MA) was defined as number of isolates that yielded falsely resistant results by disc diffusion divided by number of isolates that were susceptible by reference method (BMD). Very major error (VME) was defined as number of isolates that were falsely susceptible in disc diffusion divided by number of isolates that were resistant by reference method (BMD).

3. Results

A total of 84 CR-GNBs were collected during the study period [23 (27.4%): CRKP; 21 (25%): CRAB; 20 (23.8%): CRPA; 20 (23.8%): CREC]. The majority of patients were diagnosed with pneumonia (39.2%, 33/84), followed by BSI (33.3%, 28/84) and UTI (27.3%, 23/84). CRAB and CRPA were commonly isolated from respiratory samples whereas CREC was mostly isolated from blood samples. CRKP was isolated equally in all three samples [Fig. 1].

3.1. *In-vitro activity of cefiderocol by BMD*

Following CLSI BMD breakpoints, 76.2% (64/84) CR-GNBs were found susceptible, 5.9% (5/84) were intermediate, and 17.8% (15/84) were resistant to cefiderocol. Higher susceptibility was seen in CRE (81.4%, 35/43) [i.e. CREC and CRKP] as compared to non-fermenting GNB (70.7%, 29/41) [i.e. CRAB and CRPA]. Comparing individual groups, maximum susceptibility was seen in CRKP isolates (95.6%) and least was observed in CRAB (61.9%)[Table 1]. CRKP and CRPA isolated from patients with BSI were found 100% susceptible to cefiderocol with MIC $\leq$ 2 μg/ml. Similarly, all CRKP isolates except one isolated from urine showed good susceptibility (MICs range: 0.25–4 μg/mL). Comparing susceptible isolates of CRE, MIC distribution among CRKP was found lower than CREC. Eighty percent of CRPA were susceptible to cefiderocol except 4 isolates (two isolates each from urine and respiratory specimens). Among CREC isolates, 13 were susceptible to cefiderocol with MIC <4 μg/mL, 3 were intermediate with MIC 8 μg/mL, and 4 were resistant with MIC >32 μg/mL. All resistant CREC were isolated from patient with BSI. CRAB isolates showed highest resistance to cefiderocol (38.1%, 8/21, MIC >32 μg/mL) and majority were isolated from respiratory samples. Among CRAB trailing was seen in 33.3% (7/21) of isolates; which was ignored at time of result interpretation.

According to EUCAST, 52.3% (33/63) of total isolates were susceptible to cefiderocol. CRAB was excluded from this analysis due to lack of defined breakpoints. Amid all isolates, 82.6% of CRKP, 50% of CRPA, and 20% of CREC were susceptible to cefiderocol with MIC $\leq$2 μg/ml [Table 1]. Following EUCAST, CREC (80%) showed maximum resistance. Comparing the breakpoints by both CLSI and EUCAST, maximum concordance was observed among CRKP (87%) followed by CRPA (60%) and CREC (40%).[Table 2]. For CRAB considering the EUCAST

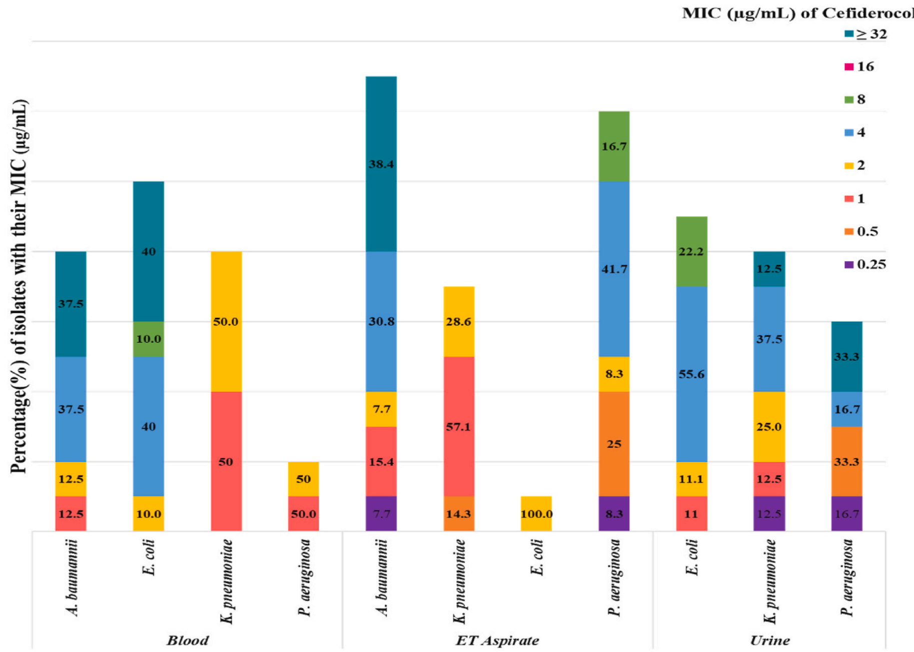

Fig. 1. Comparison of cefiderocol MIC among carbapenem-resistant GNBs isolated from different samples.

guideline, MIC values below the PK-PD breakpoint of ≤ 2 µg/ml is considered as susceptible and concordance between CLSI and EUCAST was observed in 66.6% of isolates.

3.2. *MIC_{50}/MIC_{90}*

Comparing the MIC_{50} result of cefiderocol, CRKP isolates had 2-fold lower MIC_{50} value than CRPA and 4-fold lower than CREC and CRAB isolates[Table-1]. MIC_{90} for CRKP isolates was two-fold lower than CRPA and 8 fold lower than CREC and CRAB. MIC_{50} for colistin was lowest for CRKP (0.5 µg/ml) and highest for CRPA (2 µg/ml) and MIC_{90} for colistin was equal (2 µg/ml) for all organisms except CRKP (8 µg/ml) [Table 1]. MIC_{50} and MIC_{90} for other BL/BIs were found higher as shown [Table 1].

3.3. *In-vitro susceptibility of cefiderocol by disc diffusion test*

Following the CLSI breakpoints for the disc diffusion method, 73.8% (62/84) isolates were found susceptible, 10.7% (9/84) were intermediate, and 15.5% (13/84) were resistant to cefiderocol. Maximum susceptibility was seen in CRKP (95.6%; 22/23), followed by CRPA (80%; 16/20), CRAB (61.9%; 13/21), and CREC (55%; 11/20). Following EUCAST breakpoints, only 17.5% (11/63) of isolates were susceptible to cefiderocol. The highest susceptibility was seen in CRKP (30.4%, 7/23) followed by CRPA (20%, 4/20). None of the CREC isolates was susceptible by disc diffusion according to EUCAST. In our study, 26 isolates (41.3%) were in ATU category with maximum number seen in CRPA (80%; 16/20) and a minimal of 26% (6/23) seen in CRKP and 20% (4/20) in CREC.

Using the standard definitions of agreement analysis between BMD and disc diffusion test method, CA, mE, ME, and VMEs were assessed for all the isolates. The acceptable criteria for CA is $\geq 90\%$, for mE is $\leq 10\%$ and for ME and VME $<3\%$. Following CLSI breakpoints, the CA and mE were observed within acceptable criteria; while following EUCAST breakpoints none were found within the defined limits [Table 3].

Most of BL/BLI tested were found resistant to the CR-GNB isolates in this study (piperacillin tazobactam: 98.8%, ceftazidime avibactam: 96.8%, and ceftalazone-tazobactam: 98.4%)[Table 1]. A total of 83.3% of isolates were found susceptible to colistin as compared to 76.2% for cefiderocol following CLSI breakpoints, and 52.4% following EUCAST breakpoints. Cefiderocol showed greater susceptibility in CRKP (95.7%), and CRPA (80%) isolates as compared to colistin.

Out of 84 isolates, 41 (48.8%) were detected to harbour one or both the carbapenemase genes.

The prevalence of bla_{NDM} was found higher (38.1%, 32/84) than $bla_{OXA\text{-}48}$ (3.6%, 3/84) and 7.1% (6/84) of isolates were found harbouring both the genes. 75% (24/32) of isolates harbouring bla_{NDM}, 100% (3/3) of the isolate harbouring $bla_{OXA\text{-}48}$, and 66.6%(4/6) of the isolate harbouring both the genes were found susceptible to cefiderocol [Table 4]. Complete susceptibility was seen in CRKP having one or more carbapenemase gene, indicating good action of cefiderocol against metallo-β-lactamase containing isolates.

4. Discussion

Infections by CR-GNBs are a major cause of mortality, and there is a shortage of new antibiotics for their treatment. Studies such as APEKS-cUTI, CREDIBLE-CR, and APEKS-NP led to FDA approval of cefiderocol for treating complicated UTIs and hospital-acquired bacterial pneumonia [8–10]. The European Medicines Agency (EMA) also approved it for aerobic GNB infections [11]. However, cefiderocol susceptibility has not been thoroughly studied in India. In our study, following CLSI guidelines, cefiderocol susceptibility by BMD was highest against CRKP and lowest against CRAB [12].

Various *in-vitro* studies have compared cefiderocol susceptibility in GNBs. The SIDERO-WT surveillance studies in North America and Europe (2014–2019) showed susceptibility rates of 96.7% for CRE, 99.8% for CRPA, and 94.2% for CRAB. Similarly, the SENTRY AMSP program reported over 95% susceptibility for CRE, CRPA, and CRAB13]. Nayak et al. found an overall susceptibility of 81.7% in 503 CR-GNB isolates. In our study, the susceptibility of CRKP was 95.6%, comparable to other studies, but lower for the remaining CR-GNBs.

Candel et al. found cefiderocol susceptibility rates of 77.8% for CREC, 69.2% for CRKP, 97.5% for CRPA, and 90.7% for CRAB among European isolates using EUCAST breakpoints [14]. In our study, CRKP showed the highest susceptibility by EUCAST standards. The highest concordance between EUCAST and CLSI breakpoints was seen in CRKP, followed by CRPA and CREC, aligning with existing literature [15].

Table 1
Susceptibility profile of cefiderocol and other comparators with their Minimum inhibitory concentration (MIC_{50}/ MIC_{90}).

Organism	Antibiotic	MIC_{50}; MIC_{90} (μg/ml; μg/ml)	MIC range (μg/ml)	S % (n)	I% (n)	R% (n)
E. coli **(n = 20)**	Cefiderocol	4; 32	1–32	65 % (13)	15 % (3)	20 % (4)
	Colistin	1; 2	0.25–8	NA	85 % (17)	15 % (3)
	Ceftazidime-Avibactam	>16/4; >16/4	>16/4	–	–	100 % (20)
	Ceftolozane-Tazobactam	>8/4; >8/4	>8/4	–	–	100 % (20)
	Piperacillin-Tazobactam	>32/4; >32/4	>32/4	–		100 % (20)
K. pneumoniae **(n = 23)**	Cefiderocol	1; 4	0.25–32	95.6 % (22)	–	4.35 % (1)
	Colistin	0.5; 8	0.5–8	NA	82.6 % (19)	17.4 % (4)
	Ceftazidime-Avibactam	>16/4; >16/4	<1/4 - >16/4	8.7 % (2)		91.3 % (21)
	Ceftolozane-Tazobactam	>8/4; >8/4	>8/4	–	–	100 % (23)
	Piperacillin-Tazobactam	>32/4; >32/4	>32/4	–	–	100 % (23)
P. aeruginosa **(n = 20)**	Cefiderocol	2; 8	0.25–32	80 % (16)	10 % (2)	10 % (2)
	Colistin	2; 2	0.5–4	NA	70 % (14)	30 % (6)
	Ceftazidime-Avibactam	>16/4; >16/4	>16/4	–	–	100 % (20)
	Ceftolozane-Tazobactam	>8/4; >8/4	4/4- >8/4	5 % (1)		95 % (19)
	Piperacillin-Tazobactam	>32/4; >32/4	8/4- >32/4	5 % (1)		95 % (19)
A. baumannii **(n = 21)**	Cefiderocol	4; 32	0.5–32	61.9 % (13)	–	38.1 % (8)
	Colistin	1; 2	0.5–8	NA	95.2 % (20)	4.8 % (1)
	Ceftazidime-Avibactam[a]	–	NA	–	–	–
	Ceftolozane-Tazobactam[a]	–	NA	–	–	–
	Piperacillin-Tazobactam	>32/4; >32/4	>32/4			100 % (21)

Mushtaq et al. also reported high concordance in *P. aeruginosa* and the highest discordance among Enterobacterales[16]. The discordance may be due to EUCAST low susceptibility cut-off (<2 μg/mL), categorizing isolates with MIC 2–4 μg/mL as resistant, while CLSI considers them susceptible. This raises questions about the true susceptibility of organisms with MIC 2–4 μg/mL, warranting further genomic research. Additionally, 33.3% of CRAB isolates in our study demonstrated trailing, the significance of which remains unknown and needs further investigation.

Majority of our isolates were found resistant to BL/BLIs. SIDERO-WT study observed a higher susceptibility of cefiderocol as compared to BL/BLIs for CRE and CRPA isolates [12]. Similarly, in SENTRY-AMSP trial, cefiderocol showed higher susceptibility in comparison to BL/BLIs against CRE and extensively drug-resistant *P. aeruginosa* [13]. The higher resistance to BL/BLIs in our study might be due to the increased prevalence of metallo-β-lactamase genes in these isolates.

Here, the susceptibility of cefiderocol to CRKP and CRPA was observed higher in comparison to colistin. The MIC_{50} and MIC_{90} values of cefiderocol were also found lower for these two groups in comparison to other CR-GNBs. The MIC_{50} and MIC_{90} value in SIDERO-WT studies was similar to the CRKP isolates in the present study. Considering the lower MIC of cefiderocol for CRKP and CRPA, it might be preferred over colistin for the treatment of infections caused by these organisms in Indian scenario. Further large-scale clinical trials can be planned to establish these findings.

Cefiderocol testing by BMD requires iron-depleted CAMHB medium, which is cumbersome to prepare. The availability of cefiderocol discs has simplified susceptibility testing. In our study, agreement analysis between BMD and disc diffusion using CLSI breakpoints showed complete CA for CRKP and CRAB, with acceptable minor errors for CREC and CRPA. Using EUCAST breakpoints, the highest CA was observed only in CREC. Morris et al. reported similar results, with 89% CA, 8% mE, and 3% ME using CLSI breakpoints using disc from two different pharmacies [17]. Their study also showed higher ME and VME with EUCAST breakpoints, aligning with our findings.

In our study, a higher prevalence of bla_{NDM} was seen in comparison to $bla_{OXA\text{-}48}$. Good susceptibility to cefiderocol was seen in the isolates harbouring these carbapenemase genes. *Queslati* et al. demonstrated 81% of bla_{NDM} harbouring Enterobacterales and 50% of bla_{NDM} harbouring *A. baumannii* were susceptible to cefiderocol [18].Another Indian study result showed 81.5 % susceptibility to cefiderocol for bla_{NDM} isolates and 68.7 % susceptibility for $bla_{OXA\text{-}48}$ isolates, which is in accordance with our study result [19].

Our study had a few limitations. The study was planned in a limited number of isolates detecting two carbapenemase genes. Being a tertiary care referral center, there is a possibility of more resistant isolates in our hospital. However, our study evaluated the susceptibility of cefiderocol among the CR-GNBs of Indian isolates using both CLSI and EUCAST breakpoints by BMD and Disc diffusion methods meticulously.

5. Conclusion

This study highlights the susceptibility profile of cefiderocol in Indian carbapenem-resistant isolates. Cefiderocol was found to have good action against CRKP and CRPA irrespective of the presence of bla_{NDM} and $bla_{OXA\text{-}48}$ genes. Agreement analysis results showed maximum discrepancy against CRAB isolates, which needs to be further explored with genomic studies. The MIC_{50} and MIC_{90} value of cefiderocol was observed much lower in CRKP and CRPA, which might have better

Table 2
Comparative evaluation of susceptibility profile of CR-GNBs following CLSI and EUCAST breakpoints.

CLSI (S ≤ 4 μg/mL; I 8 μg/mL; R ≥ 16 μg/mL)	E U C A S T (S ≤ 2 μg/mL; R > 2 μg/mL)								
		K.pneumoniae **(n = 23)** No.(%)		***P. aeruginosa*** **(n = 20)** No. (%)		***E.coli*** **(n = 20)** No.(%)		***A.baumannii*** **(n = 21)** No.(%)	
		S	**R**	**S**	**R**	**S**	**R**	**S**	**R**
	S	19 (82.6 %)	3 (13 %)	10 (50 %)	6 (30 %)	4 (20 %)	9 (45 %)	6 (28.6 %)	7 (33.3 %)
	I				2 (10 %)		3 (15 %)		
	R		1 (4.4 %)		2 (10 %)		4 (20 %)		8 (38.1 %)

Table 3
Agreement analysis of Cefiderocol for various CR-GNBs: Disc diffusion vs Microbroth dilution.

Organism	CLSI (%)				EUCAST (%)			
	CA	mE	ME	VME	CA	mE	ME	VME
E.coli (n = 20)	90 % (18)	10 % (2)	–	–	80 % (16)	–	20 % (4)	–
K.pneumoniae(n = 23)	100 % (23)	–	–	–	47.8 % (11)	–	52.2 % (12)	–
P.aeruginosa (n = 20)	90 % (18)	10 % (2)	–	–	60 % (12)	–	35 % (7)	5 % (1)
A.baumannii (n = 21)	100 % (21)	–	–	–	71.4 % (15)	–	–	40 % (6)

Table 4
Susceptibility of cefiderocol in CR-GNBs with/without *bla*$_{NDM}$ and *bla*$_{oxa-48}$ genes.

Cefiderocol susceptibility	*K.pneumoniae* (23)			*E.coli* (20)			*P.aeruginosa* (20)			*A.baumannii* (21)			Total (84)		
	S (22)	I (0)	R (1)	S (13)	I(3)	R(4)	S (16)	I(2)	R(2)	S(13)	I (0)	R (8)	S (64)	I (5)	R(15)
Only NDM	100 % (4/4)	0	0	70 % (7/10)	10 % (1/10)	20 % (2/10)	87.5 % (7/8)	0	12.5 % (1/8)	60 % (6/10)	0	40 % (4/10)	75 % (24/32)	3.1 % (1/32)	21.9 % (7/32)
Only OXA-48	100 % (3/3)	0	0	0	0	0	0	0	0	0	0	0	100 % (3/3)	0	0
Both NDM and OXA-48	100 % (3/3)	0	0	33.3 % (1/3)	33.3 % (1/3)	33.3 % (1/3)	0	0	0	0	0	0	66.7 % (4/6)	16.6 % (1/6)	16.6 % (1/6)
Both negative	92.3 % (12/13)	0	7.7 % (1/13)	71.4 % (5/7)	14.3 % (1/7)	14.3 % (1/7)	75 % (9/12)	16.7 % (2/12)	8.3 % (1/12)	63.6 % (7/11)	0	36.4 % (4/11)	76.7 % (33/43)	7 % (3/43)	16.3 % (7/43)

therapeutic potential for Indian patients. Multicentric studies with larger sample sizes will be helpful to check its therapeutic efficacy in Indian settings.

CRediT authorship contribution statement

Akansha Didwania: Writing – review & editing, Writing – original draft, Visualization, Project administration, Investigation, Formal analysis, Data curation. **Sarita Mohapatra:** Writing – review & editing, Visualization, Supervision, Project administration, Funding acquisition, Conceptualization. **Deepak Kocher:** Writing – review & editing, Investigation, Formal analysis, Data curation. **Hitender Gautam:** Writing – review & editing, Supervision. **Priyanka Kumari:** Investigation, Data curation. **Arvind Kumar:** Supervision, Project administration, Conceptualization. **Manish Soneja:** Supervision, Project administration. **Naval K. Vikram:** Supervision, Project administration. **Seema Sood:** Writing – review & editing, Supervision. **Benu Dhawan:** Writing – review & editing, Supervision. **Bimal Kumar Das:** Writing – review & editing, Supervision. **Rama Chaudhry:** Writing – review & editing, Supervision. **Manoranjan Mahapatra:** Supervision, Project administration. **Arti Kapil:** Writing – review & editing, Supervision.

Ethic approval and consent to participate

This study was approved by the institute ethics committee [All India Institute of Medical Sciences, New Delhi (reference number: IEC-760/November 12, 2021,RP-31/2021)]. Informed consent was obtained from all the subjects involved in the study.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Not applicable.

List of Abbreviations:

MIC	Minimum Inhibitory Concentration
GNB	Gram-negative bacteria
CR-GNB	Carbapenem-resistant Gram-Negative Bacilli
CRAB:	Carbapenem-Resistant *Acinetobacter baumannii*
CRPA	Carbapenem-Resistant *Pseudomonas aeruginosa*
CREC	Carbapenem-Resistant *Escherichia coli*
CRKP:	Carbapenem-Resistant *Klebsiella pneumoniae*
CRE	Carbapenem-Resistant Enterobacteriales
BMD	Broth microdilution
DD	Disc Diffusion
CLSI	Clinical and Laboratory Standards Institute
EUCAST	European Committee on Antimicrobial Susceptibility Testing
BL/BLI	β-lactam/β-lactamase inhibitors
ESBL:	Extended-spectrum beta-lactamases
BSI	Bloodstream infection
UTI:	Urinary Tract Infection
MALDI-TOF	Matrix Assisted Laser Desorption Ionization-Time of Flight
AST	Antimicrobial susceptibility testing
CA-MHB	Cation-adjusted Mueller Hinton broth
MIC_{50}	Minimum inhibitory concentration of 50 percentile
MIC_{90}	Minimum inhibitory concentration of 90 percentile
CA-MHA	Cation-adjusted Mueller Hinton agar
ATU	Area of Technical Uncertainty
PCR	Polymerase chain reaction
CA	Categorical agreement
mE	Minor error
MA	Major error
VME	Very major error
PK-PD	Pharmcokinetics-pharmacodynamics
EMA	European Medicines Agency
ICMR	Indian Council of Medical Research

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.ijmmb.2025.100793.

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